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ADR-0185: Hive-Mind Consensus Cli Retirement

Context and Problem Statement

[ADR-0184](#) closed (2026-05-18, Wave 6a) with all 6 per-strategy bodies live in `forks/agentdb/src/archivist/handlers/hive-mind/consensus/<strategy>.ts`. The Wave 6a exit-gate test verifies zero `pending` stubs in the agentdb handler tree and audit-entry count = mutation count. Acceptance baseline maintained at 672/681/0/9 across all 6 waves.

The cli `hive-mind_consensus` handler at `forks/ruflo/v3/@claude-flow/cli/src/mcp-tools/hive-mind-tools.ts` (lines 1984-2919, ~870 LoC handler body + ~50 LoC schema/header) is now **100% redundant** with the agentdb per-strategy handlers. Both surfaces work correctly; they are dual-write. ADR-0184 Wave 6 envisioned flipping the cli to a thin `archivist.dispatch('hive-mind_consensus', payload)` wrapper, but scope measurement during Wave 6 execution revealed the cli flip itself is a substantial focused refactor:

Concern	Approximate scope
<code>buildConsensusResponse(state, action, strategy)</code> — reconstructs ~20 telemetry fields per action from POST-dispatch state	~300-400 LoC standalone
Try/catch error-reshape (5 typed-error branches → <code>{action, error, ...}</code> envelope; 3 re-throw branches)	~30 LoC
Helper-set cleanup (10 cli-local helpers: zero external callers confirmed; need to split dead-after-flip from still-callable-by-response-builder)	~200 LoC delete + ~100 LoC keep
Cli test re-validation (existing assertions in <code>__tests__/mcp-tools-deep.test.ts</code> for response shapes, error classes, byzantine detection, gossip rounds, CRDT merges)	full re-run + per-assertion review
Total	~1500 LoC of focused refactor work

This spin-out follows the [ADR-0181](#) → [ADR-0184](#) precedent (peel out a single concern when the closure plan would otherwise bloat). ADR-0184 Wave 6a delivered the agentdb-side correctness gates; ADR-0185 picks up the cli retirement as a focused next step.

Decision Drivers

- `feedback-no-fallbacks` — every cli soft- `{action, error}` return must map cleanly to either a typed throw (reshaped post-dispatch) or a re-throw (preserving cli's pre-flip contract). No silent coerce.
- `feedback-no-value-judgements-on-features` — default to WIRE. The cli flip unifies the consensus surface; the dual-write state is structurally redundant and should be collapsed.
- Response-shape parity is its own correctness gate.** The cli's existing response objects span ~20 telemetry fields per action × 4 actions × 6 strategies. Each field must round-trip exactly through `buildConsensusResponse(...)` or downstream MCP callers break.
- One-shot 1500-LoC was the risk that motivated this spin-out.** ADR-0184 Wave 6 attempted a one-shot retire; scope measurement triggered a stand-down. ADR-0185 takes the time to do it incrementally with proper verification.
- ADR-0184 Wave 6a exit-gate test (zero `pending` stubs, audit-equals-mutation count) verifies the agentdb surface is COMPLETE independent of cli retirement. ADR-0185 does not block any future archivist work.

- `feedback-singleton-frozen-state-desync` (ADR-0184 Open Follow-up #4): cli tests that touch `getProcessHiveMindStore()` may need `CLAUDE_FLOW_CWD` env-var isolation. ADR-0185 carries the responsibility for this guard.

Considered Options

- **A. 1500-LoC one-shot retire.** Rejected — this is the one-shot risk that motivated the spin-out. ADR-0184 Wave 6 stood down to avoid it.
- **B. Per-action incremental flip** (propose first → vote → status → list). Each action lands as its own commit; cli's strategy fan-out gets thinner with each commit. Cli tests can be re-validated per action.
- **C. Per-strategy + per-action matrix** (24 commits across 6 strategies × 4 actions). Maximum granularity; minimum per-commit risk; substantial coordination overhead.
- **D. Hybrid: response-builder-first then per-action flip.** Land `buildConsensusResponse(...)` as its own commit (pure additive; called by an "experimental" code path that mirrors the existing handler's output). Then flip each action one at a time. Verification via parallel-strategy harness: drive identical inputs through cli + archivist, diff responses.

Decision Outcome

Chosen: **Option D (hybrid)** — land `buildConsensusResponse(...)` additively first; verify response parity via a parallel-strategy harness; then flip each action one at a time with the harness still running as a regression guard.

Option A (one-shot retire) is rejected: the 1500-LoC scope is the exact risk that motivated this spin-out from ADR-0184 Wave 6.

Option B (per-action flip without prior response-builder) is rejected: each per-action flip would need response-builder logic `INLINE` in that action's branch, duplicating work across 4 actions. Option D extracts the logic `ONCE` in Wave 1 and reuses it through Waves 2-5.

Option C (24-cell matrix) is rejected: 6 strategies × 4 actions × per-cell verification is overhead disproportionate to risk. The response-builder handles strategy-conditional fields internally; per-action flip is the natural granularity for cli changes.

Option D sequences risk so that the response-builder bug (if any) surfaces at `LAND` time (Wave 1, additive, no behaviour change), not at `FLIP` time (Waves 2-5, where a bug would correlate with the cli flip and obscure trace). Per-action flip then becomes a 1-line dispatcher swap per action, with the response-builder already proven correct against the cli's pre-flip baseline.

Consequences

- Good — incremental verification at each action's flip. Per-action commit revert is surgical.
- Good — the response-builder is a pure read function over `POST`-dispatch state; no mutation, no I/O. Trivially testable in isolation.
- Bad — Option D requires more wall-clock than Option A (multiple commits + multiple release gates) IF Option A would have worked one-shot.
- Neutral — cli `crdt-types.ts` deletion is a separate cli-housekeeping concern. ADR-0185 may delete the cli copy iff the cli's `buildConsensusResponse` is the last consumer, OR leave it as an orphaned export.

Confirmation

- `npm run release` is the gate for each per-action commit.
- **Parallel-strategy verification harness:** drive identical inputs through cli (pre-flip) AND archivist (post-flip via test-only direct call) and diff responses; the diff must be empty for every action × strategy combination.
- Per-action invariants must throw on falsy per `feedback-no-fallbacks` — no silent pass branches.
- Final gate: cli `hive-mind_consensus` handler body is a thin dispatch wrapper; helper-set cleanup landed; cli `crdt-types.ts` deletion decision documented.

Architecture

- `buildConsensusResponse(action, strategy, proposalId, state, input) → ConsensusResponse` at NEW FILE `forke/ruflo/v3/@claude-flow/cli/src/mcp-tools/hive-mind-consensus-response.ts` (~300-400 LoC). Pure function. Locates the proposal in `state.consensus.pending` first, falls back to `state.consensus.history`.

Signature:

```
export function buildConsensusResponse(
  action: 'propose' | 'vote' | 'status' | 'list',
  strategy: 'bft' | 'raft' | 'quorum' | 'weighted' | 'gossip' | 'crdt',
  proposalId: string,
  state: HiveMindState,
  input: HiveMindConsensusPayload,
): ConsensusResponse;
```

Return type `ConsensusResponse` is a **TypeScript discriminated union** on `action`: `ProposeResponse` | `VoteResponse` | `StatusResponse` | `ListResponse`. Each member carries strategy-conditional optional fields (gossip-only: `gossipRound`, `gossipBound`, `gossipExhausted`; crdt-only: `crdtState`, `crdtVerdict`, `crdtApprovers`, `crdtVoteCount`, `crdtExpectedVoters`, `crdtTimedOut`; weighted-only: `weight-sum` subfield; byzantine-only: `byzantineVoters`). The union shape is the formal cli response spec; downstream MCP-caller documentation can cite it directly. (This resolves Open Follow-up #1.)

- **Cli `hive-mind_consensus` handler** at `forks/ruflo/v3/@claude-flow/cli/src/mcp-tools/hive-mind-tools.ts` — post-retirement shape is a thin wrapper (~40-50 LoC, down from ~870):
 1. Pre-mint `proposalId` for `action === 'propose'` (mirrors `task_create` precedent at `mcp-tools/task-tools.ts:127-160`). Other actions take `proposalId` from `payload`.
 2. `await archivist.dispatch('hive-mind_consensus', { ...payload, proposalId })` for write actions (`propose` | `vote` | `status`), `archivist.dispatchRead('hive-mind_consensus', payload)` for `list`. Typed-overload form per ADR-0181 Phase 5 `ToolPayloadMap`.
 3. Try/catch with `instanceof` discrimination (error classes re-exported from `@sparkleideas/agentdb/archivist` per ADR-0184 Wave 6a barrel):
 - **Reshape to `{action, error: <e.message>, ...} envelope`** (5): `RaftTermCollisionError`, `RaftVoteChangeError`, `DuplicateVoteError`, `VoterIdRequiredError`, `ProposalNotFoundError`.
 - **Re-throw** (3, per cli's pre-flip contract): `MissingQueenForWeightedConsensusError`, `WorkerAlreadyFailedError`, `ProposalAlreadyFailedError`.
 4. Re-read state via `archivist.dispatchRead('hive-mind_status', { proposalId })`.
 5. return `buildConsensusResponse(action, strategy, proposalId, state, input)`.
- **Helper-set cleanup** in `forks/ruflo/v3/@claude-flow/cli/src/mcp-tools/hive-mind-tools.ts`:
 - **DELETE** — mutation-only, dead-after-flip (verify zero external callers via `grep -rE` per helper before delete): `detectByzantineVoters`, `tryResolveProposal`, `maybeAdvanceGossipRoundOnTimeout`, `selectGossipTargets`, `reconcileFailedFromStatusKeys`, `workerMetaFor`. Approximately ~200 LoC across the 6 functions.
 - **KEEP in cli** — pure-read, called by `buildConsensusResponse`: `calculateRequiredVotes`, `weightedTally`, `settleCheckGossip`, `gossipFanout`. **Vendor-from-agentdb is rejected** — response-building is a cli concern; cross-package import for pure-read helpers adds coupling that doesn't pay back. (This resolves Open Follow-up #5.)
 - **KEEP cli `crdt-types.ts`** — `buildConsensusResponse` calls `LWWRegister.from(...).value()` / `ORSet.from(...).elements()` / `GCounter.from(...).value()` for CRDT telemetry. Cross-package import (`@sparkleideas/agentdb/archivist/handlers/hive-mind/consensus/_crdt-types`) would add coupling without benefit. (This resolves Open Follow-up #3.)
- **Parallel-strategy verification harness** at NEW FILE `forks/ruflo/v3/@claude-flow/cli/__tests__/hive-mind-consensus-parity.test.ts` (~400 LoC):
 - Drives 24 inputs (4 actions × 6 strategies) — `propose` × {`bft`, `raft`, `quorum`, `weighted`, `gossip`, `crdt`}, same for `vote` / `status` / `list` — through BOTH cli's pre-flip handler AND archivist's dispatched handler. (For the `list` action, "across 6 strategies" is degenerate; harness uses 1 input there → 19 distinct cells in practice, expanded with state variants to ≥24 effective scenarios.)
 - Diffs responses field-by-field via:
 - `crdtSemanticEqual()` (already in `agentdb` per ADR-0184 Wave 5 `_shared.ts`) for CRDT subfields — handles `ORSet`'s internal-ordering issue.

- Plain deep-equal for other fields.
- **Determinism strategy** (resolves Open Follow-up #2):
 - Clock: `vi.useFakeTimers()` + `vi.setSystemTime(<fixed epoch>)` per test, advance via `vi.advanceTimersByTime()` for ADR-0131 timeout-driven transitions.
 - `proposalId` : pre-mint via FNV-1a + mulberry32 seed (same pattern ADR-0184 Wave 5 used). Cli's `propose` action accepts the pre-minted id via payload.
 - ORSet tagging: cli's ORSet uses `Math.random()` for tag generation; harness stubs `Math.random` with a seeded mulberry32 sequence in `beforeEach`.
 - Per-test cleanup: `vi.restoreAllMocks()` in `afterEach`.
- **Singleton isolation** (resolves Open Follow-up #4): each test uses `CLAUDE_FLOW_CWD` env-var injection (per ADR-0183 A0 swarm pattern) + `vi.stubEnv('HOME', ...)` to scope cli singletons (`getProcessHiveMindStore()`, etc.) per-test. Vitest `@claude-flow/memory` externalization (also per ADR-0183 A0) may be required.
- **Harness lifecycle**: ACTIVE during Waves 1-5 as the regression guard. Decision in Wave 6: KEEP as permanent regression guard once cli is fully dispatched. Cost is ~400 LoC of test code; benefit is detecting any future drift between cli + archivist response shapes.

Execution Plan

6 sequential waves under Option D hybrid. Each wave gates on `npm run release` (default acceptance — heavy gate only at Wave 6 close-out). Per-wave commit on `forks/ruflo` main with descriptive message naming the action flipped + harness verdict; no Co-Authored-By trailer.

Wave	Scope	Files	Net LoC	Exit gate
1	Land <code>buildConsensusResponse(...)</code> additively + parallel-strategy verification harness. Cli <code>hive-mind_consensus</code> handler UNCHANGED — harness verifies the response-builder produces the cli's pre-flip shape exactly.	NEW: <code>cli/src/mcp-tools/hive-mind-consensus-response.ts</code> (350 LoC including <code>ConsensusResponse</code> union); <code>cli/__tests__/hive-mind-consensus-parity.test.ts</code> (400 LoC). NO CHANGE: <code>cli/src/mcp-tools/hive-mind-tools.ts</code> .	+750	<code>npm run release</code> passes; harness asserts zero response diffs across all 24 cells (<code>{propose,vote,status,list} × {bft,raft,quorum,weighted,gossip,crdt}</code> cli handler body unchanged (grep-asserted).
2	Flip <code>propose</code> action. Cli's propose branch: pre-mint <code>proposalId</code> , <code>archivist.dispatch('hive-mind_consensus', { ...payload, proposalId })</code> , try/catch with instanceof discrimination on 8 typed errors, re-read state via <code>dispatchRead('hive-mind_status')</code> , return <code>buildConsensusResponse(...)</code> . Other 3 action branches stay on cli fan-out.	MODIFIED: <code>cli/src/mcp-tools/hive-mind-tools.ts</code> (propose branch only, ~80 LoC → ~25 LoC).	-55	<code>npm run release</code> passes; harness asserts zero propose-response diff across all 6 strategies; existing <code>cli/__tests__/mcp-tools-deep.test.ts</code> propose-related assertions pass; ADR-0184 audit-equals-mutation invariant preserved.

Wave	Scope	Files	Net LoC	Exit gate
3	Flip <code>vote</code> action. Same shape as Wave 2 but for vote branch. The vote branch is heavier in cli (~200 LoC) because it carries strategy-specific tally/equivocation logic — that logic is now in agentdb per ADR-0184; cli just dispatches + builds response.	MODIFIED: <code>cli/src/mcp-tools/hive-mind-tools.ts</code> (vote branch).	-180	<code>npm run release</code> passes; harness asserts zero vote-response diff; existing vote-related assertions pass (Byzantine equivocation detection still surfaces correctly via response shape).
4	Flip <code>status</code> action. ADR-0131 inline-timing decision (from ADR-0184 Wave 4) means status is a WRITE action; cli uses <code>archivist.dispatch</code> not <code>dispatchRead</code> . Harness must drive timeout-driven transitions via <code>vi.advanceTimersByTime()</code> and verify the auto-status-transition fields (<code>statusJustTransitioned</code> , <code>timedOut</code> , <code>settled</code>) round-trip.	MODIFIED: <code>cli/src/mcp-tools/hive-mind-tools.ts</code> (status branch).	-270	<code>npm run release</code> passes; harness asserts zero status-response diff across all 6 strategies INCLUDING ADR-0131 timeout-driven transitions; gossip-strategy <code>gossipExhausted</code> flag (ADR-0184 Wave 4) round-trips.
5	Flip <code>list</code> action. Pure read — cli uses <code>archivist.dispatchRead('hive-mind_consensus', { action: 'list', ... })</code> .	MODIFIED: <code>cli/src/mcp-tools/hive-mind-tools.ts</code> (list branch).	-45	<code>npm run release</code> passes; harness asserts zero list-response diff. Cli <code>hive-mind_consensus</code> handler body is now ~40-5 LoC of dispatch logic only.
6	Helper-set cleanup + close-out. Delete the 6 mutation-only helpers; verify zero external callers via <code>grep -rE</code> per helper. Keep the 4 pure-read helpers + cli <code>crdt-types.ts</code> per §Architecture decisions. Author close-out report + ADR-0185 amendment. Decide harness lifecycle (recommended: KEEP permanent).	MODIFIED: <code>cli/src/mcp-tools/hive-mind-tools.ts</code> (delete 6 helpers). NEW: <code>docs/council/ADR-0185-close-out-report.md</code> , ADR-0185 <code>### Amendment: Close-out</code> section.	-200	<code>npm run release</code> passes (default + <code>ACCEPTANCE_HEAVY=1</code> heavy gate to mirror ADR-0181/0184 close-out posture); zero unreachable helpers in <code>mcp-tools/hive-mind-tools.ts</code> ; cli <code>hive-mind_consensus</code> handler body ~40-50 LoC; ADR-0185 close-out report + amendment land in a ruflo-patch commit.

Net LoC delta: +750 (Wave 1 additive) – 750 (Waves 2-6 deletions) = 0 net, but the cli handler body shrinks –870 → –50 LoC. The 6 mutation-only helpers (200 LoC) are net deleted; the response-builder + harness (~750 LoC) is net added.

Estimated release-gate cycles: 6 (one per wave). Plus 1 heavy gate at Wave 6 close-out (matches ADR-0181/0184 close-out parity).

Per-wave revert posture: each wave's commit is surgical (single action branch flip in Waves 2-5; pure additive in Wave 1; deletion-only in Wave 6). If a wave's harness asserts non-empty diff, the action under flip is the immediate suspect — revert that wave's commit cleanly, trace the diff, re-attempt. Per `feedback-trace-before-hypothesis`.

Swarm sizing: can run as a queen-as-implementer per the ADR-0181/0184 pattern, OR as `/swarm-advanced` per-action workers if the in-process Agent-spawn constraint is lifted by the time ADR-0185 starts. The work is naturally serial (each wave depends on the prior); parallel-worker fan-out doesn't add value within a wave.

Open Follow-ups

- 1. **Response-shape-spec-capture — RESOLVED in §Architecture:** `ConsensusResponse` is a TypeScript discriminated union on `action`, with `ProposeResponse` | `VoteResponse` | `StatusResponse` | `ListResponse` members carrying strategy-conditional optional fields. The union shape IS the formal spec.
- 2. **Parallel-strategy-verification-harness — RESOLVED in §Architecture:** harness at `cli/__tests__/hive-mind-consensus-parity.test.ts`; determinism via `vi.useFakeTimers()` + `vi.setSystemTime()`, `FNV-1a` + `mulberry32` `proposalId` seeding, `Math.random` stubbing for `ORSet` tags, `crdtSemanticEqual()` for CRDT subfield diff.
- 3. **Cli-crdt-types-ts-deletion — RESOLVED in §Architecture:** KEEP cli copy. `buildConsensusResponse` calls its constructors for CRDT telemetry; cross-package import would add coupling without benefit.
- 4. **feedback-singleton-frozen-state-desync —** partially resolved in §Architecture (harness uses `CLAUDE_FLOW_CWD` + `vi.stubEnv('HOME', ...)`). **Remaining concern:** Vitest `@claude-flow/memory` externalization may be required (precedent: ADR-0183 A0 swarm vitest config). Verify during Wave 1 harness authoring.
- 5. **Cli-helpers — vendor-vs-import-for-pure-read-functions — RESOLVED in §Architecture:** KEEP cli copies of `calculateRequiredVotes`, `weightedTally`, `settleCheckGossip`, `gossipFanout`. Response-building is a cli concern; cross-package import for pure-read helpers adds coupling that doesn't pay back.
- 6. **Harness lifecycle decision at Wave 6 close-out.** Current §Architecture lean: KEEP as permanent regression guard (cost ~400 LoC of test code; benefit catches future drift between cli + archivist response shapes). Alternative: retire iff cli is so thin post-Wave-6 that drift is impossible. Decision binding at Wave 6 close-out commit; default is KEEP.
- 7. **ADR-0184 Wave 6a gossipExhausted? flag —** added to `ConsensusProposal` in agentdb Wave 4. Cli's `buildConsensusResponse` reads it via the dispatched-response payload; harness must include a gossip-exhausted state variant to verify the flag round-trips.
- 8. **Cli-test assertion-set refresh —** `cli/__tests__/mcp-tools-deep.test.ts` currently asserts response shapes from the pre-flip cli handler. Post-flip the shapes are produced by `buildConsensusResponse`; assertion targets are nominally unchanged but should be reviewed per wave to confirm no assertion drifts behind the abstraction.

Amendments

Amendment: Close-out — ADR-0185 implementation complete (2026-05-18)

ADR-0185 is implemented and closed. The cli `hive-mind_consensus` handler is now a thin dispatch envelope; the strategy fan-out is gone.

Execution structure — 6 waves (Wave 2 split into 2a + 2b by DA mandate):

Wave	Scope	Fork SHA
W1	Additive <code>buildConsensusResponse</code> + parity harness (26 cells) + ruflo-patch wrapper gate	c81831164, 425710868, b43e039f3, eeb26cd7f
W2a	Harness shape-contract pivot (DA-mandated split)	a20732b98
W2b	Flip propose action to <code>archivist.dispatch</code>	7e06c8390
W3	Flip vote action	ca596e932
W4	Flip status action (includes ADR-0131 timeouts)	f4b9ffe12
W5	Flip list action + delete <code>withHiveStoreLock</code> wrapper	84defa083
W6	Delete 3 dead helpers + close-out (this amendment)	bf72ed19a

Cumulative cli LoC reduction: handler 870 → 127 LoC + 3 dead helpers (128 LoC) deleted. Total cli surface shrink across Waves 1-6 ≈ -870 LoC. Response-builder + parity harness add ~1150 LoC of test/projection code in their place.

Acceptance trajectory: 672/681/0/9 sustained across all 6 default-gate releases (patches 190 → 192 → 193 → 194 → 197 → 199 → 202 → 203). Heavy gate cleared at close-out.

Helper-cleanup correction (Wave 6, DA Axis 1): the pre-Wave-6 plan listed 6 helpers for deletion. Pre-emptive `grep` on `src/` + `__tests__` confirmed only 3 were truly unreachable. The other 3 — `selectGossipTargets`, `reconcileFailedFromStatusKeys`, `workerMetaFor` — have live callers in `hive-mind_status` (a separate cli tool, explicitly deferred per the ADR's scope) plus active test-invariant refs in `mcp-tools-deep.test.ts`. Per `feedback-no-squelch-tests` they were preserved.

Source-grep brittleness lesson: 4 of 6 acceptance-test failures across the 6 waves were `assert.match(src, ...)` strings searching for identifiers that had relocated from cli to agentdb per-strategy handlers. Each pattern needed surgical migration (cli → agentdb dispatcher case-arm + per-strategy handler case-arm). Wave 7+ work should convert remaining `assert.match(src, ...)` assertions to behavioural tests so the corpus survives future relocations.

Parity-harness lifecycle (Wave 6, DA Axis 6): the parity harness (29 cells) + the ruflo-patch wrapper gate (3 cells) are permanent regression guards per ADR-0185 §Architecture. No future wave may delete them — they lock in cli/agentdb response-shape parity for the lifetime of the dispatch pattern.

More Information

- [ADR-0184 close-out report](#) — the parent close-out that records the cli retirement spin-out.
- [ADR-0184: Hive-Mind Consensus Handler Port](#) — the parent ADR; all 6 per-strategy bodies live in agentdb per Wave 1-6a.
- [ADR-0181: Archivist Runtime Activation §Closure plan amendment](#) — the closure-plan-amendment precedent for sequencing the work.
- [ADR-0183: Memory Write-Path Unification close-out](#) — the spin-out precedent for peeling a single concern out of a parent closure plan.
- `forks/ruflo/v3/@claude-flow/cli/src/mcp-tools/hive-mind-tools.ts` lines 1984-2919 — the 926-LoC cli handler this ADR retires.
- `forks/ruflo/v3/@claude-flow/cli/src/mcp-tools/task-tools.ts` lines 127-160 — the `task_create` precedent for cli-flip pattern (pre-mint id, dispatch, post-dispatch re-read).
- `forks/agentdb/src/archivist/handlers/hive-mind/consensus/_shared.ts` — agentdb's vendored cli helpers + 8 typed error classes (re-exported via `archivist/index.ts` per ADR-0184 Wave 6a).
- `forks/agentdb/test/archivist/handlers/exit-gate.test.ts` — the agentdb-side exit-gate test that verifies handler completeness independent of cli retirement.